



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 2 • STANDARD 6.NOS.1

# Interpret Division of Fractions

Flagship

Lesson 2-1-flagship · Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## EVIDENCE LAB MISSION

### **The Fraction Vault**

You are the lead forensic analyst at the Reyes Detective Agency. A 3-foot strip of evidence tape must be cut into exact  $\frac{1}{4}$ -foot sections before it can be logged — and the vault that holds the case file only opens once you know how many pieces that makes. Interpret fraction division and crack the case.

## My Notes

Level 3 Enrichment



### TODAY'S OBJECTIVES

**Content Objective:** I can use models to interpret what it means to divide by a fraction.

**Language Objective:** I can explain a fraction division using the words dividend, divisor, quotient, reciprocal, and unit fraction.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Dividend

Divisor

Quotient

Reciprocal

Unit fraction



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

In  $6 \div 1/2$ , the number 6 being divided is the .

2

In  $6 \div 1/2$ , the number  $1/2$  we divide by is the .

3

The answer to a division problem is the .

4

The fraction you get by flipping the numerator and denominator is the

.

5

A fraction with a numerator of 1, like  $1/5$ , is a .

## Write About the Math

### Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

#### EXPLAIN

**On your fraction bar, how many  $\frac{1}{4}$ -foot sections fit into one whole foot?  
Point to where you see the pieces?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

### 2. Plan Your Math Words

Check at least two words you will use.

☐ **Dividend** — The number you are splitting up in a division problem.  
*Dividendo — El número que estás repartiendo en una división.*

☐ **Divisor** — The number you split by in a division problem.  
*Divisor — El número entre el que divides en una división.*

☐ **Quotient** — The answer when you divide.  
*Cociente — La respuesta cuando divides.*

☐ **Unit fraction** — A fraction with 1 on top, like  $\frac{1}{4}$ .  
*Fracción unitaria — Una fracción con 1 arriba, como  $\frac{1}{4}$ .*

☐ **Reciprocal** — A fraction turned upside down.  
*Recíproco — Una fracción volteada de arriba abajo.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**In one whole foot I see \_\_\_\_ sections of  $\frac{1}{4}$  each.**

*En un pie completo veo \_\_\_\_ secciones de  $\frac{1}{4}$  cada una.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** Because each piece is smaller than a whole foot, 3 feet makes more than 3 pieces.

**Evidence:** Students reason that since  $\frac{1}{4}$  is less than 1, more than 3 pieces fit, so the quotient is greater than the dividend.

**Reasoning:** This evidence connects the definition of dividend to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **In one whole foot I see \_\_\_\_ sections of  $\frac{1}{4}$  each.**  
*En un pie completo veo \_\_\_\_ secciones de  $\frac{1}{4}$  cada una.*
- **So 3 feet hold \_\_\_\_ pieces because 3 times \_\_\_\_ equals \_\_\_\_.**  
*Entonces 3 pies tienen \_\_\_\_ pedazos porque 3 por \_\_\_\_ es \_\_\_\_.*

---

---

---

---

#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **I know because \_\_\_\_.**  
*Lo sé porque \_\_\_\_.*
- **So \_\_\_\_.**  
*Entonces \_\_\_\_.*

---

---

---

---

## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

---

---

---

---

---

---

## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*

- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*

- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*

- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

---

## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

---

## Answer Key & Teacher Guide

1. **Notes 1:** Dividend
2. **Notes 2:** Divisor
3. **Notes 3:** Quotient
4. **Notes 4:** Reciprocal
5. **Notes 5:** Unit fraction
6. **Watch for this mistake:** *A common mistake in Interpret Division of Fractions is assuming division always makes the answer smaller — a rule that holds for whole numbers but breaks down when dividing by a fraction less than 1. For example, in  $6 \div 1/2$ , a student expects a smaller answer and calculates  $6 \times 1/2 = 3$ , when the correct answer is  $6 \div 1/2 = 12$ , because 12 halves fit inside 6 wholes. The same error shows up in word problems: if a recipe needs  $1/2$  cup of flour per batch and you have 6 cups, a student might reason 'dividing means less' and answer 3 batches instead of the correct 12 batches. Before you submit, ask: am I dividing by a fraction smaller than 1? If so, the answer should be BIGGER than the number I started with.*
7. **Try It 1:** A.  $8 \div 1/4 = 32$  sections — *Each foot holds 4 fourths, so 8 feet hold  $8 \times 4 = 32$  sections. The matching expression is  $8 \div 1/4 = 32$ .*
8. **Try It 2:** A. The number of  $1/3$ -quart beakers she can fill, which is 15 —  *$5 \div 1/3$  asks how many  $1/3$ -quart beakers fit into 5 quarts. Each quart holds 3 thirds, so  $5 \times 3 = 15$  beakers.*